

Sample Problems for Test 1

Problem 1. Let $\mathbf{A} = \begin{bmatrix} -1 & 1 & 2 \\ 3 & 2 & -2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & -2 \\ -1 & 4 \end{bmatrix}$.

- a) Which one of the products $\mathbf{AB}^t$ and $\mathbf{A}^t\mathbf{B}$ is possible? Explain.
- b) Compute the product from a) that is possible.

Problem 2. Consider the 2×2 matrices

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 2a_{11} + 3a_{21} & 2a_{12} + 3a_{22} \\ 5a_{11} - 2a_{21} & 5a_{12} - 2a_{22} \end{bmatrix}.$$

Find a 2×2 matrix $\mathbf{B}$ such that $\mathbf{BA} = \mathbf{C}$.

Problem 3. Let $\mathbf{A} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 4 \\ -1 & 5 & -2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix}$.

- a) Find the $\mathbf{L} - \mathbf{U} - \mathbf{P}$ factorization of $\mathbf{A}$ using the strategy of partial pivoting.
- b) Using the $\mathbf{L} - \mathbf{U} - \mathbf{P}$ factorization from a) to solve $\mathbf{Ax} = \mathbf{b}$

Problem 4. Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 5 \\ 0 & 0 & 1 \end{bmatrix}$.

- a) Which one of $\mathbf{A}$ and $\mathbf{B}$ is invertible? Explain.
- b) Find the inverse matrix of the one from a) that is invertible.

Problem 5. Let $\mathbf{A} = \begin{bmatrix} 1 & -2 & & & \\ -1 & 4 & 3 & & \\ & 2 & 4 & -1 & \\ & & 2 & -4 & 4 \\ & & & -2 & 6 \end{bmatrix}$.

- a) Is $\mathbf{A}$ column diagonally dominant?
- b) Find the $\mathbf{L} - \mathbf{U}$ factorization of the tridiagonal matrix $\mathbf{A}$ using the special procedure for tridiagonal matrices.

Problem 6. Let $\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 2 \end{bmatrix}$ and $\mathbf{B} = \mathbf{I}_{5 \times 5}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 3 \\ 0 \\ 1 \\ 3 \end{bmatrix}$.

- a) Find $\mathbf{u}$ and $\mathbf{v}$ such that $\mathbf{A} = \mathbf{B} - \mathbf{uv}^t$.
- b) Use a) and the sherman-Morrison algorithm to solve $\mathbf{Ax} = \mathbf{b}$.

Problem 7. True or False

- a) Let $\mathbf{A}$ and $\mathbf{B}$ be 2×2 matrix. If $\mathbf{AB} = 0$, then either $\mathbf{A} = 0$ or $\mathbf{B} = 0$.
- b) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrix. Then $(\mathbf{AB})^{-1} = \mathbf{A}^{-1}\mathbf{B}^{-1}$.
- c) Let $\mathbf{u}$ and $\mathbf{v}$ be two nonzero column n -vectors. If the matrix $\mathbf{I} - \mathbf{uv}^t$ is nonsingular, then $\mathbf{v}^t\mathbf{u}=1$.

Problem 8. How many flops does it take to compute the product of an upper triangular $n \times n$ matrix and a column n -vector if multiplication by zero is not executed? Explain.